

```
clc;

clear;

close all;

% Parameters

fs = 1000;      % Sampling frequency

t = 0:1/fs:1;   % Time vector (1 second)

TBW = 600;      % Total Bandwidth (kHz)

Users = 3;      % Number of users

CBW = TBW / Users; % Bandwidth per user

% Carrier frequencies

fc = [100 300 500];

% Message signal

m = sin(2*pi*10*t);

% FDMA signals

s1 = m .* cos(2*pi*fc(1)*t);

s2 = m .* cos(2*pi*fc(2)*t);

s3 = m .* cos(2*pi*fc(3)*t);

% Plot transmitted signals

subplot(2,1,1);

plot(t, s1, 'r', t, s2, 'g', t, s3, 'b');

grid on;

title('FDMA Channel Allocation');

xlabel('Time (s)');

ylabel('Amplitude');

legend('User 1','User 2','User 3');

% Plot channel bandwidth allocation

subplot(2,1,2);
```

```

bar(1:Users, CBW*ones(1,Users));

grid on;

title('Channel Bandwidth per User');

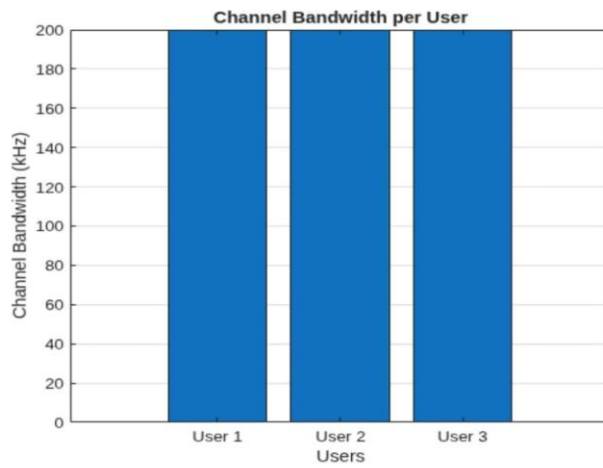
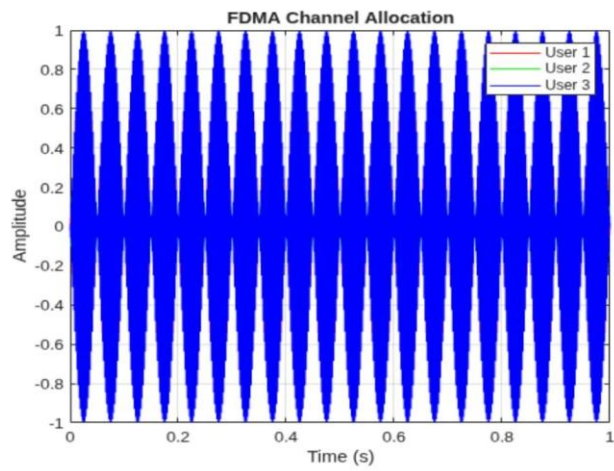
xlabel('Users');

ylabel('Channel Bandwidth (kHz)');

xticks(1:Users);

xticklabels({'User 1','User 2','User 3'});

```



```
clc; clear; close all;
```

```
BW = 200; % Channel Bandwidth (kHz)
```

```
Users = 3;
```

```
M = 2; % BPSK Modulation
```

```
DataRate = BW*log2(M); % Data Rate (kbps)
```

```
Rate = DataRate*ones(1,Users);
```

```
disp(' User Data Rate (kbps)')
```

```
disp([(1:Users)' Rate'])
```

```
figure
```

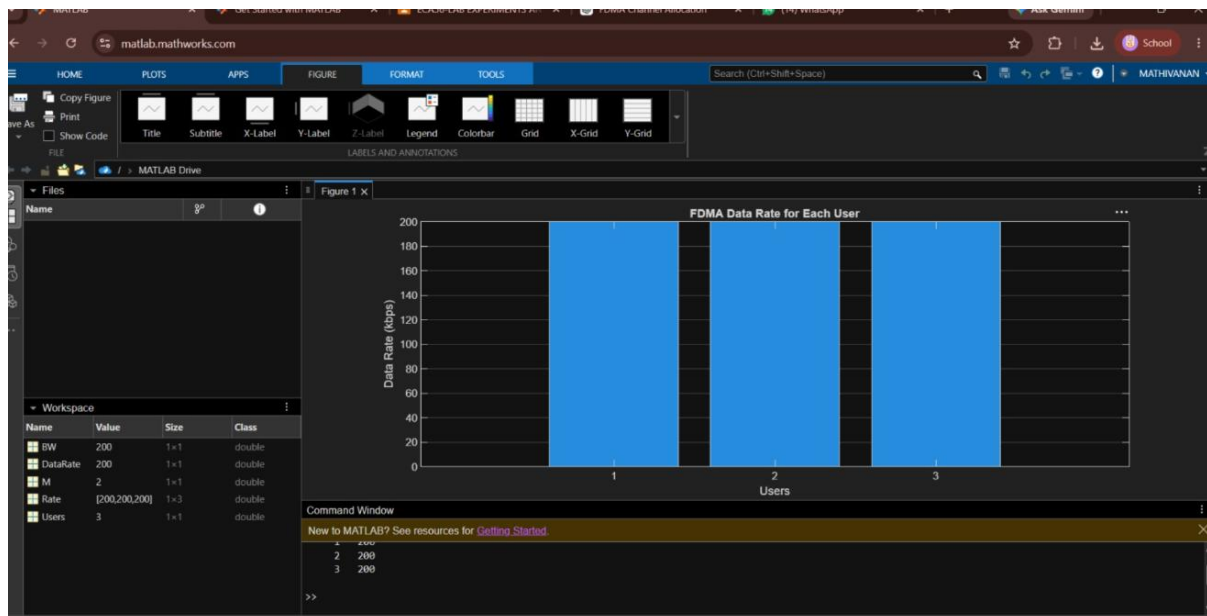
```
bar(1:Users,Rate)
```

```
xlabel('Users')
```

```
ylabel('Data Rate (kbps)')
```

```
title('FDMA Data Rate for Each User')
```

```
grid on
```



```
clc; clear; close all;
```

```
Ps = 1; % Signal Power (W)
```

```
Pn = [0.1 0.05 0.01]; % Noise Power (W)
```

```
Users = 1:3;
```

```
SNR = 10*log10(Ps./Pn); % SNR in dB
```

```
disp('User Noise Power(W) SNR(dB)')
```

```
disp([Users' Pn' SNR'])
```

```
figure
```

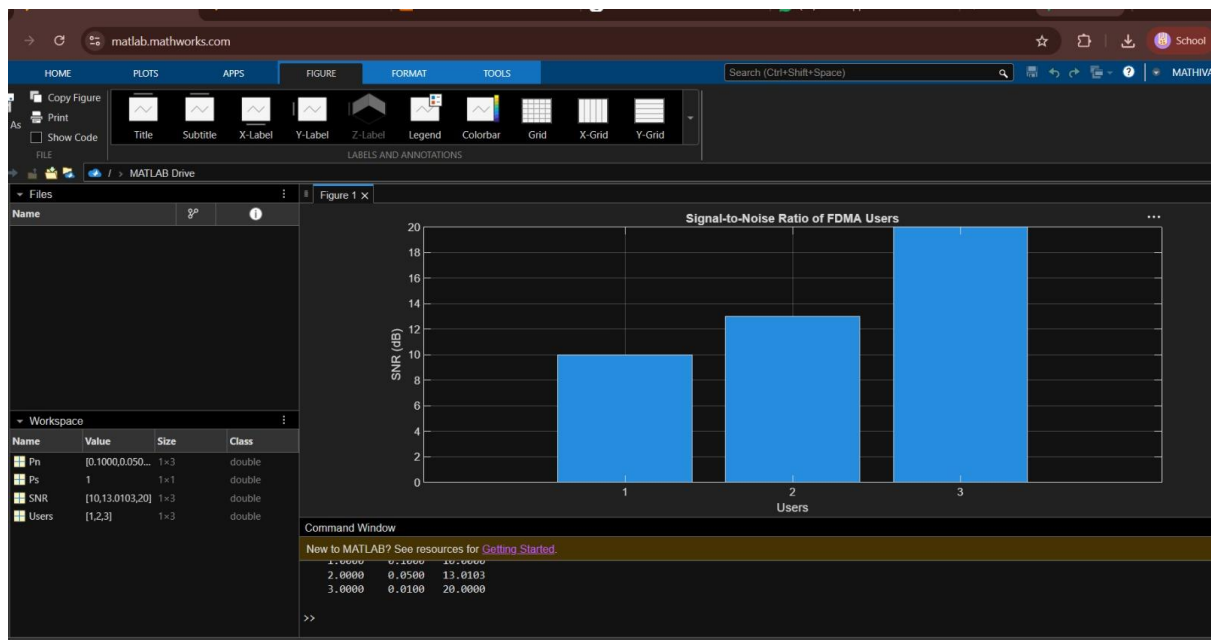
```
bar(Users,SNR)
```

```
xlabel('Users')
```

```
ylabel('SNR (dB)')
```

```
title('Signal-to-Noise Ratio of FDMA Users')
```

```
grid on
```

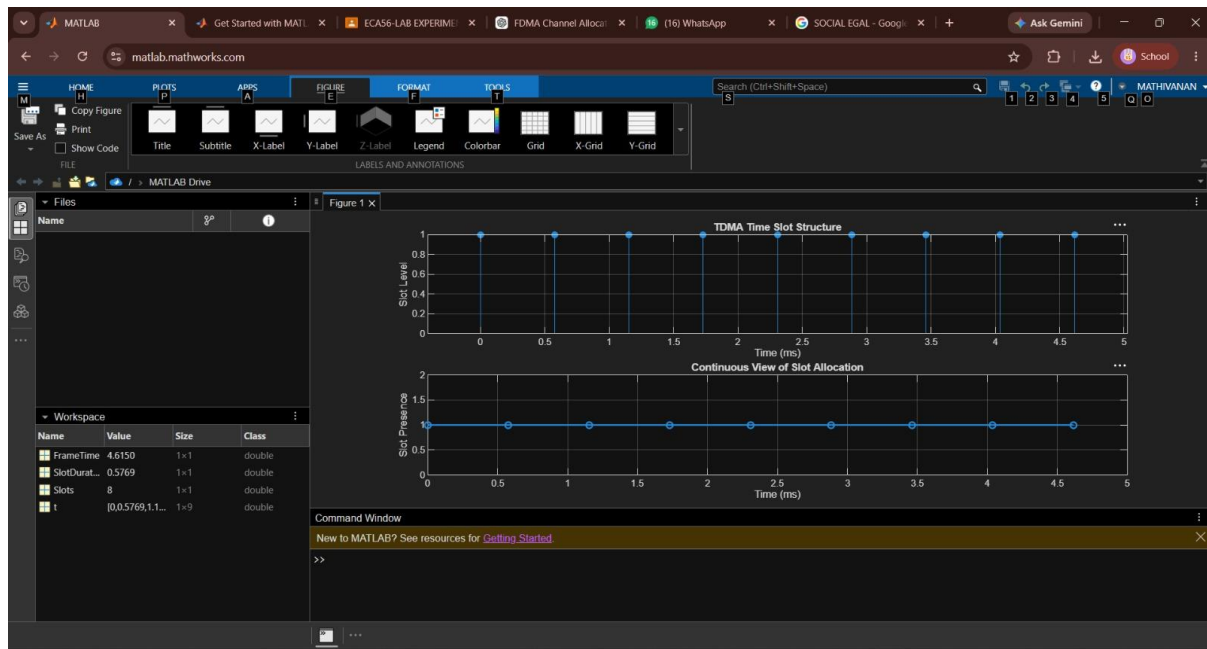


```
clc; clear; close all;

FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;

t = 0:SlotDuration:FrameTime;

figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```



```
clc; clear; close all;
```

```
TotalSlots = 8;
```

```
AllocatedSlots = 6;
```

```
FreeSlots = TotalSlots - AllocatedSlots;
```

```
Utilization = (AllocatedSlots/TotalSlots)*100;
```

```
disp('Frame Utilization (%)')
```

```
disp(Utilization)
```

```
figure
```

```
% Graph 1: Simple Line Plot (SAFE - no bar function)
```

```
subplot(2,1,1)
```

```
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
```

```
title('GSM Frame Utilization (Allocated vs Free)')
```

```
xlabel('Slot Type Index (1=Allocated, 2=Free)')
```

```
ylabel('Number of Slots')
```

```
grid on
```

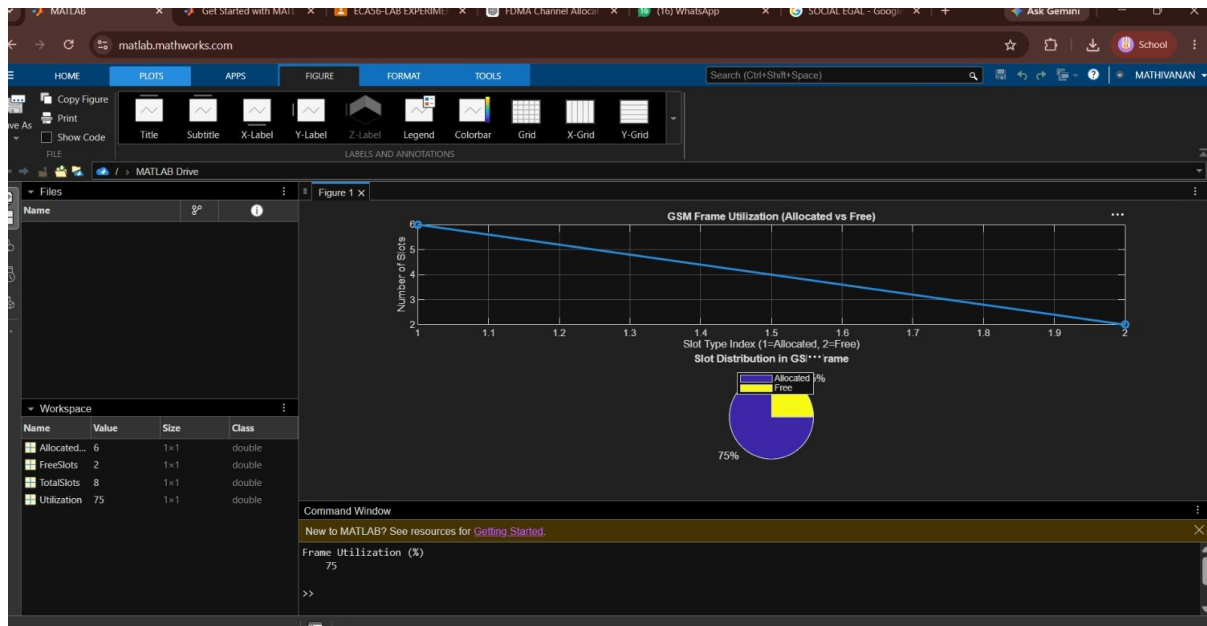
```
% Graph 2: Pie Chart (SAFE)
```

```
subplot(2,1,2)
```

```
pie([AllocatedSlots FreeSlots])
```

```
title('Slot Distribution in GSM Frame')
```

```
legend('Allocated','Free')
```



```
clc;
```

```
clear;
```

```
close all;
```

```
% Parameters
```

```
Users = 1:8;    % Total users
```

```
TotalSlots = 5;    % Available slots
```

```
% Initialize allocation
```

```
Alloc = zeros(1, length(Users));
```

```
Alloc(1:TotalSlots) = 1;    % Allocate first 5 users
```

```
% Blocked users
```

```
Blocked = 1 - Alloc;
```

```
% Display allocation table
```

```
disp('User Wise Slot Allocation');
```

```

T = table(Users', Alloc', ...
    'VariableNames', {'User', 'Status'});
disp(T);

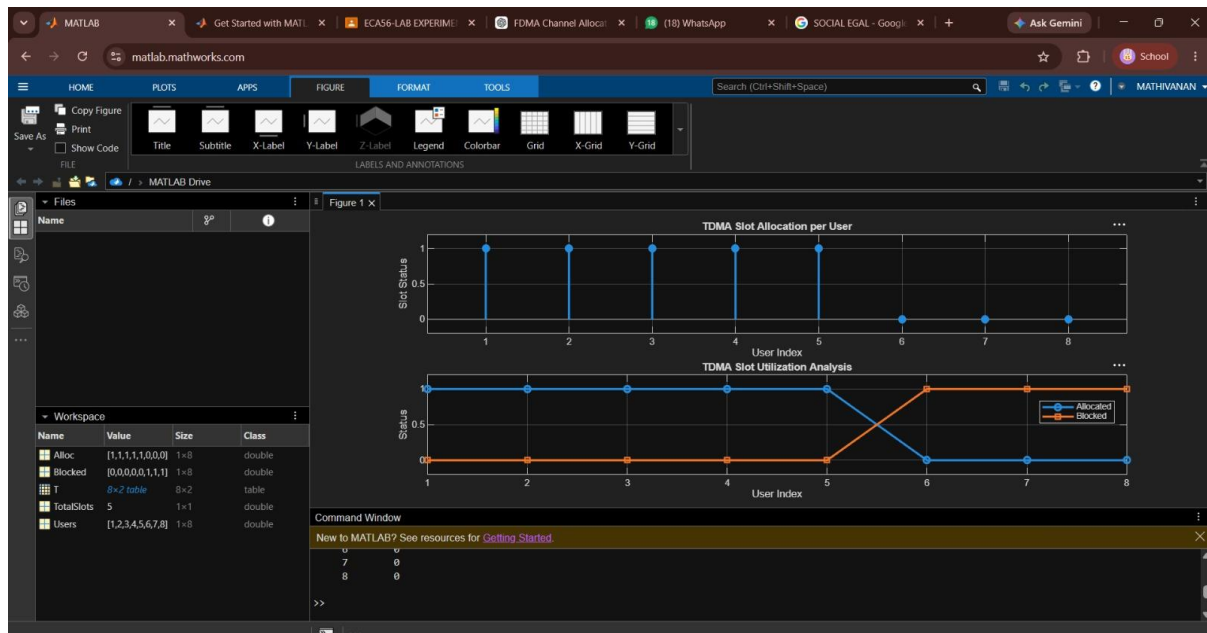
% Create figure
figure;

%----- Graph 1 -----%
subplot(2,1,1);
stem(Users, Alloc, 'filled', 'LineWidth', 2);
title('TDMA Slot Allocation per User');
xlabel('User Index');
ylabel('Slot Status');
ylim([-0.2 1.2]);
xticks(Users);
grid on;

%----- Graph 2 -----%
subplot(2,1,2);
plot(Users, Alloc, '-o', 'LineWidth', 2);
hold on;
plot(Users, Blocked, '-s', 'LineWidth', 2);
hold off;

title('TDMA Slot Utilization Analysis');
xlabel('User Index');
ylabel('Status');
legend('Allocated', 'Blocked', 'Location', 'best');
ylim([-0.2 1.2]);
xticks(Users);
grid on;

```



clc; clear; close all;

d = 0.1:0.1:5; % Distance (km)

f = [900 3500 28000]; % Frequency (MHz)

PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);

PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);

PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(d,PL1,'b','LineWidth',2); hold on

plot(d,PL2,'r','LineWidth',2)

plot(d,PL3,'k','LineWidth',2)

title('Propagation Loss vs Distance')

xlabel('Distance (km)')

ylabel('Loss (dB)')

grid on

% ----- Graph 2 (NO legend used to avoid error) -----

```

subplot(2,1,2)

Freq = [900 3500 28000];

Loss = [PL1(10) PL2(10) PL3(10)];

plot(Freq, Loss, '-o', 'LineWidth', 2)

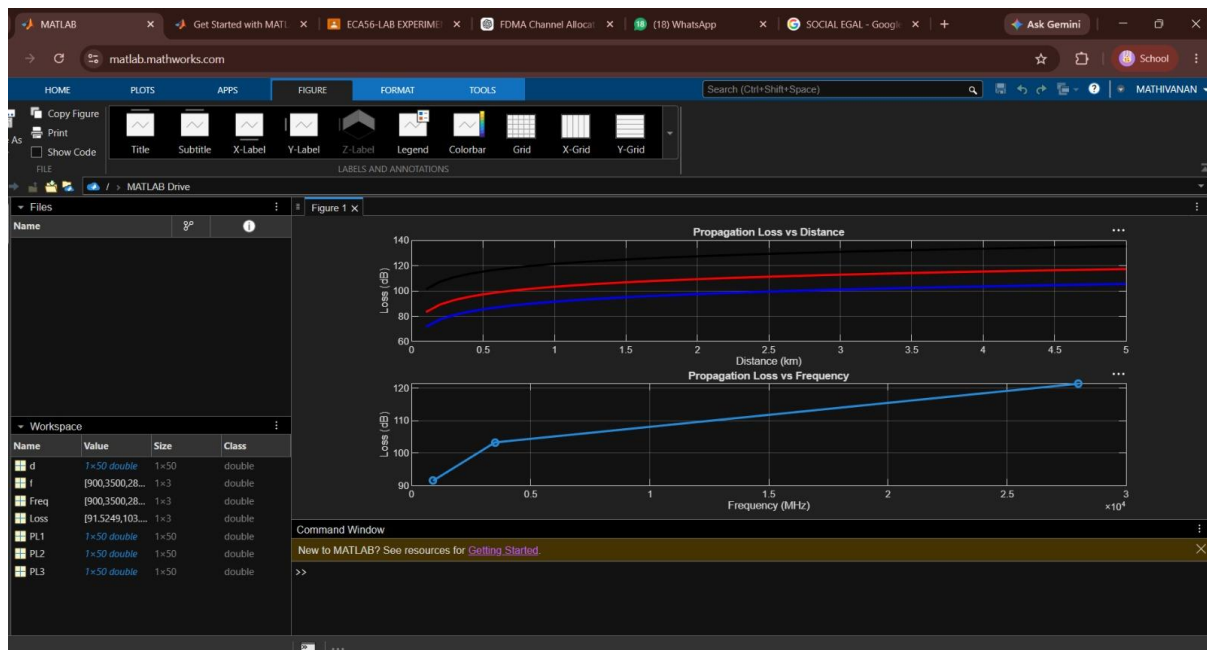
title('Propagation Loss vs Frequency')

xlabel('Frequency (MHz)')

ylabel('Loss (dB)')

grid on

```



```

clc; clear; close all;

d = 2; % fixed distance (km)

f = [900 3500 28000]; % MHz

PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

% ----- Graph 1: Frequency vs Loss -----

subplot(2,1,1)

plot(f, PL, '-o', 'LineWidth', 2)

title('Propagation Loss vs Operating Frequency')

xlabel('Frequency (MHz)')

ylabel('Propagation Loss (dB)')

grid on

```

```
% ----- Graph 2: Band Comparison -----
```

```
subplot(2,1,2)
```

```
bar(PL)
```

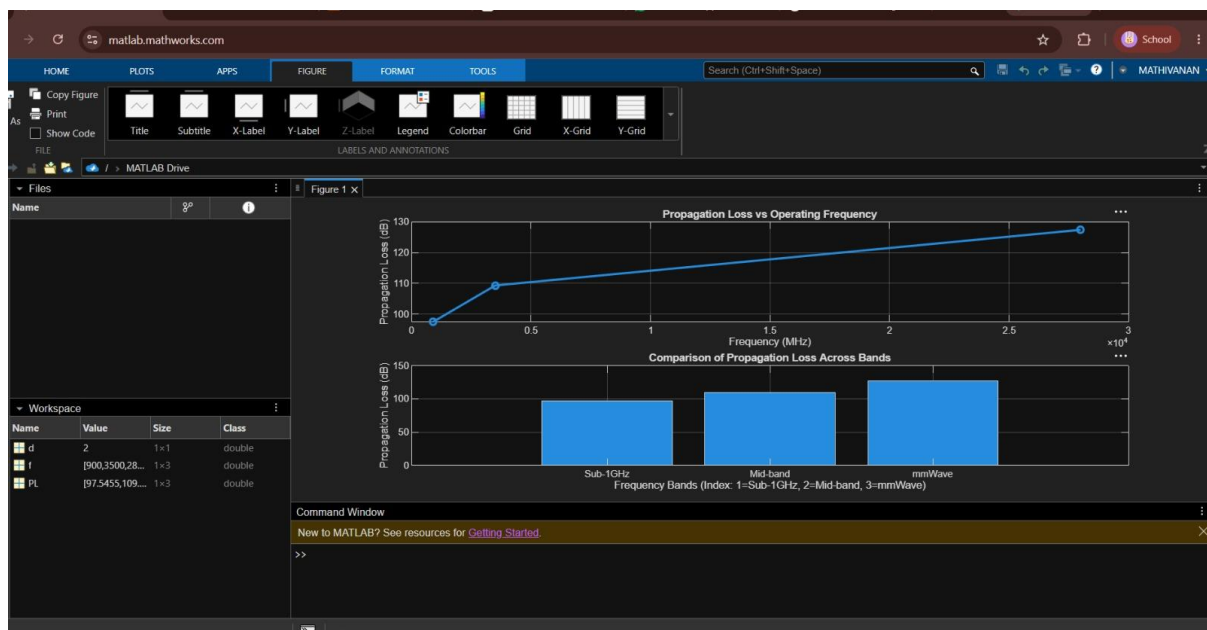
```
title('Comparison of Propagation Loss Across Bands')
```

```
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
```

```
ylabel('Propagation Loss (dB)')
```

```
xticklabels({'Sub-1GHz','Mid-band','mmWave'})
```

```
grid on
```



```
clc; clear; close all;
```

```
% Frequency bands (MHz)
```

```
f = [900 3500 28000];
```

```
% Distance (km)
```

```
d = 2;
```

```
% Propagation Loss (FSPL)
```

```
PL = 32.44 + 20*log10(f) + 20*log10(d);
```

```
figure
```

```
% ----- Graph 1: Simple Line Plot (SAFE) -----
```

```
subplot(2,1,1)
```

```

plot(f, PL, '-o', 'LineWidth', 2)

title('Propagation Loss vs Operating Frequency')

xlabel('Frequency (MHz)')

ylabel('Propagation Loss (dB)')

grid on

% ----- Graph 2: Safe Comparison Plot -----

subplot(2,1,2)

plot(1:3, PL, '-s', 'LineWidth', 2)

title('Band-wise Propagation Loss Comparison')

xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')

ylabel('Propagation Loss (dB)')

grid on

```

